MODULE 1
TRANSDUCERS INTRODUCTION – Science of Measurement & Instrumentation

Science of Measurement and Instrumentation

Functional Elements of Measurement Systems

- Scientists, engineers and civilians of all nations use a vast range of instruments to perform their measurements.
- These instruments may range from simple objects such as ruler scales and stopwatches to electron microscopes and particle accelerators used by scientists and engineers.
- An instrument is as a device or a system which is designed to maintain a functional relationship between prescribed properties of physical variables being measured.
- It provides the means of communication to a human observer or the operator of a machine or equipment.
- The performance of an instrument of a measurement system is usually described in terms of a set of its static and dynamic characteristics.

Science of Measurement and Instrumentation

Functional Elements of Measurement Systems

- We begin by defining a process as a system which generates information. Examples are a chemical reactor, a jet fighter, a gas platform, a submarine, a car, a human heart, and a weather system.
- information variables which are commonly generated by processes: thus a car generates displacement, velocity and acceleration variables, and a chemical reactor generates temperature, pressure and composition variables.
- then define the observer as a person who needs this information from the process. This could be the car driver, the plant operator or the nurse.
- The purpose of the measurement system is to link the observer to the process

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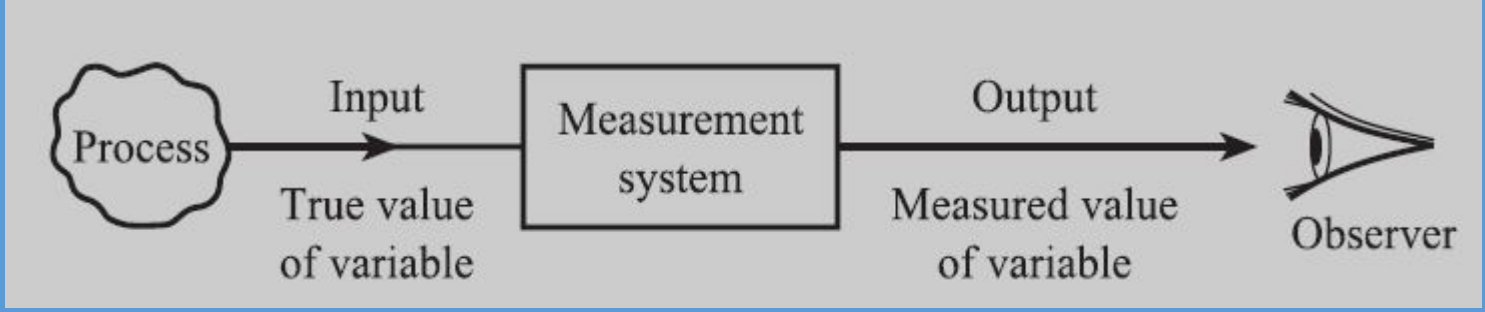
Functional Elements of Measurement Systems

- Will refer this picture which illustrates purpose of measurement systems



- We can now refer to the information variable as a measured variable. The input to the measurement system is the true value of the variable; the system output is the measured value of the variable.
- In an ideal measurement system, the measured value would be equal to the true value. The accuracy of the system can be defined as the closeness of the measured value to the true value.

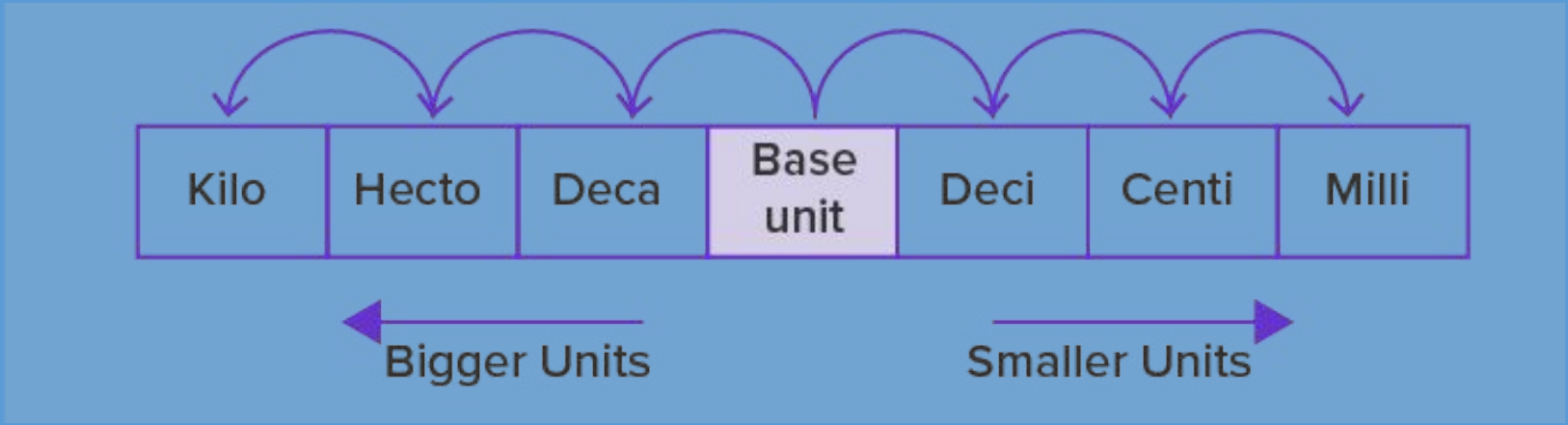
Functional Elements of Measurement Systems

- A perfectly accurate system is a theoretical ideal and the accuracy of a real system is quantified using measurement system error E, where

  E = measured value − true value

  E = system output − system input

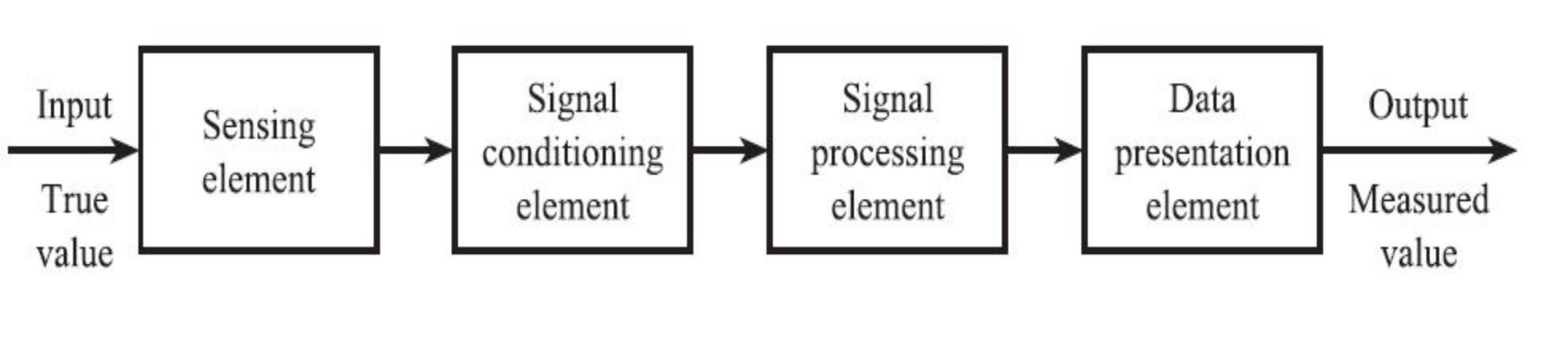
- Thus if the measured value of the flow rate of gas in a pipe is 11.0 m3/h and the true value is 11.2 m3/h, then the error E = −0.2 m3/h.
- If the measured value of the rotational speed of an engine is 3140 rpm and the true value is 3133 rpm, then E = +7 rpm.
- Error is the main performance indicator for a measurement system



Kilo	**Hecto**	**Deca**	Gram	**Decigram**	**Centi**	**Milli**
1000 kg	100 hg	10 dag	g	1/10 dg	1/100 cg	1/1000 mg

Functional Elements of Measurement Systems:
Structure of measurement systems

The measurement system consists of several elements or blocks. It is possible to identify four types of element, although in a given system one type of element may be missing or may occur more than once. The four types are shown in Figure and can be defined as follows.



Functional Elements of Measurement Systems-
Structure of measurement system

Sensing element

This is in contact with the process and gives an output which depends in some way on the variable to be measured. Examples are:

- Thermocouple where millivolt e.m.f. depends on temperature
- Strain gauge where resistance depends on mechanical strain
- Orifice plate where pressure drop depends on flow rate.

If there is more than one sensing element in a system, the element in contact with the process is termed the primary sensing element, the others secondary sensing elements.

Functional Elements of Measurement Systems-
Structure of Measureement System

Signal conditioning element

This takes the output of the sensing element and converts it into a form more suitable for further processing, usually a d.c. voltage, d.c. current or frequency signal.

Examples are:

- Deflection bridge which converts an impedance change into a voltage change
- Amplifier which amplifies millivolts to volts
- Oscillator which converts an impedance change into a variable frequency voltage.

Functional Elements of Measurement Systems-Structure of Measurement System

Signal processing element

- This takes the output of the conditioning element and converts it into a form more suitable for presentation. Examples are:
- Analogue-to-digital converter (ADC) which converts a voltage into a digital form for input to a computer
- Computer which calculates the measured value of the variable from the incoming digital data.

Typical calculations are:

- Computation of total mass of product gas from flow rate and density data
- Integration of chromatograph peaks to give the composition of a gas stream
- Correction for sensing element non-linearity.

Functional Elements of Measurement Systems-
Structure of Measurement System

Data presentation element

- This presents the measured value in a form which can be easily recognized by the observer.

  Examples are:

- Simple pointer–scale indicator
- Chart recorder
- Alphanumeric display
- Visual display unit (VDU).

Examples of measurement systems

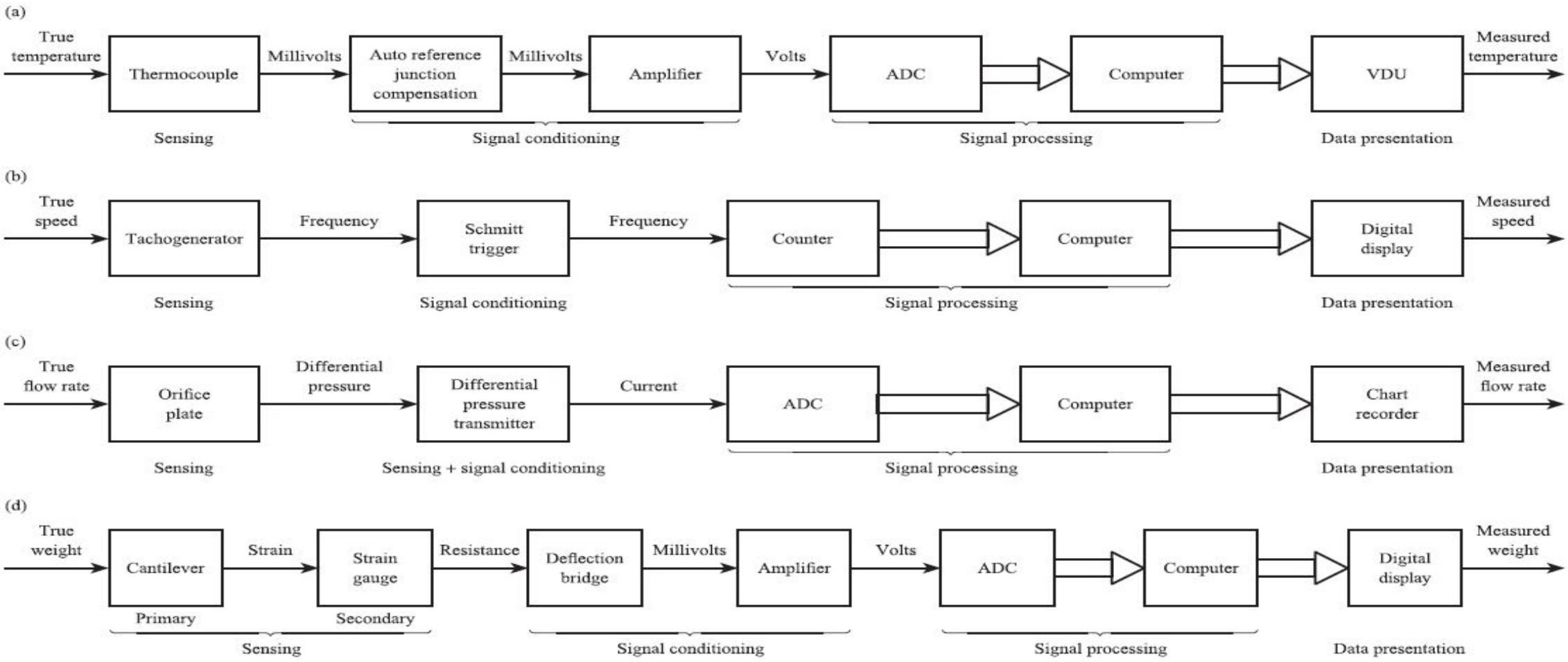


Figure 1.3 Examples of measurement systems.

Examples of measurement systems

Fig1.3(a) shows a temperature system with a thermocouple sensing element, this gives a millivolt output. Signal conditioning consists of a circuit to compensate for changes in reference junction temperature, and an amplifier. The voltage signal is converted into digital form using an analogue-to-digital converter, the computer corrects for sensor non-linearity, and the measured value is displayed on a VDU.

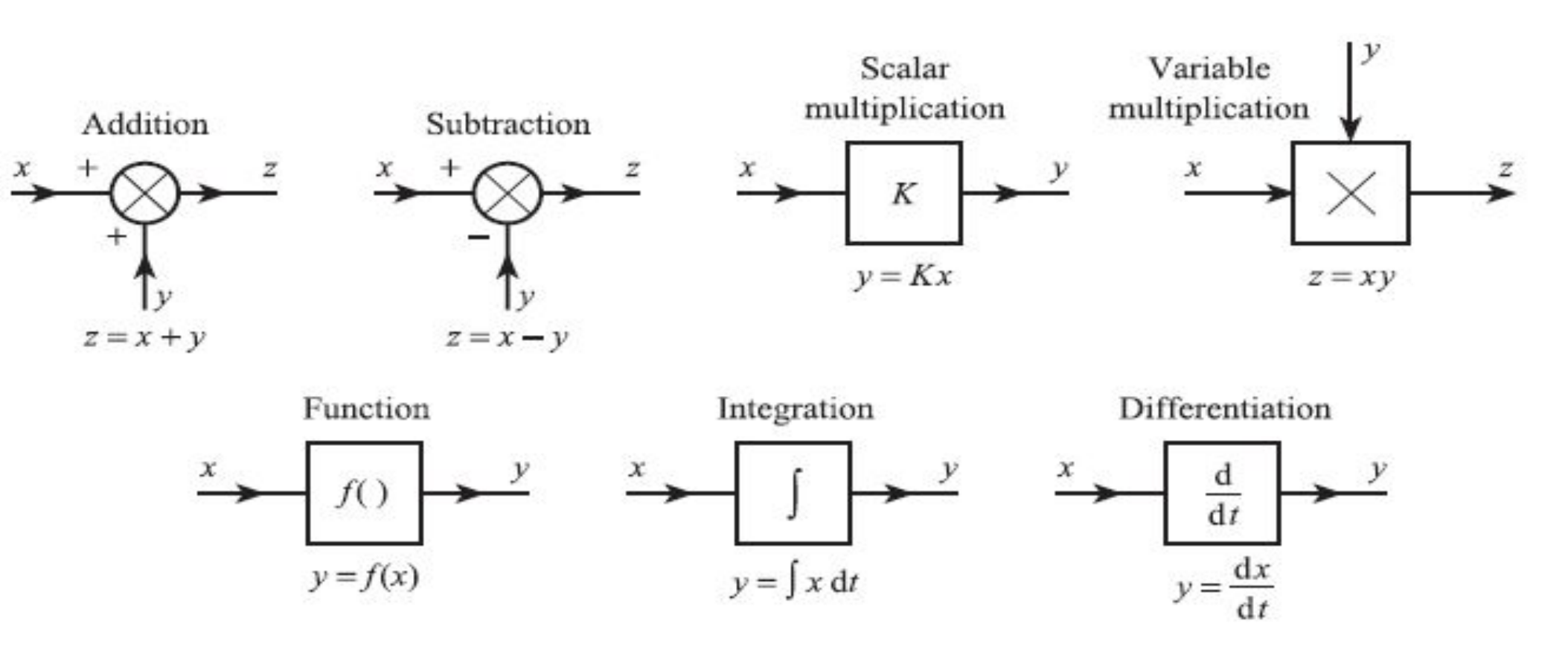
In Figure 1.3(b) the speed of rotation of an engine is sensed by an electromagnetic tachogenerator which gives an a.c. output signal with frequency proportional to speed. The Schmitt trigger converts the sine wave into sharp-edged pulses which are then counted over a fixed time interval. The digital count is transferred to a computer which calculates frequency and speed, and the speed is presented on a digital display.

Examples of measurement systems

The flow system of Figure 1.3(c) has an orifice plate sensing element; this gives a differential pressure output. The differential pressure transmitter converts this into a current signal and therefore combines both sensing and signal conditioning stages. The ADC converts the current into digital form and the computer calculates the flow rate, which is obtained as a permanent record on a chart recorder.

The weight system of Figure 1.3(d) has two sensing elements: the primary element is a cantilever which converts weight into strain; the strain gauge converts this into a change in electrical resistance and acts as a secondary sensor. There are two signal conditioning elements: the deflection bridge converts the resistance change into millivolts and the amplifier converts millivolts into volts. The computer corrects for non-linearity in the cantilever and the weight is presented on a digital display.

Block diagram symbols



principles of sensing and transduction

Transducer

- **Definition:** The device which converts the one form of energy into another is known as the transducer. The process of conversion is known as transduction. The conversion is done by sensing and transducing the physical quantities like temperature, pressure, sound, etc.

Sensor: a device which detects or measures a physical property and records, indicates, or otherwise responds to it.

principles of sensing and transduction

Needs of Transducer

It is quite difficult to determine the exact magnitude of the physical forces like temperature, pressure, etc. But if the physical force is converted into an electrical signal, then their value is easily measured with the help of the meter. The transducers convert the physical forces into an electrical signal which can easily be handled and transmitted for measurement.

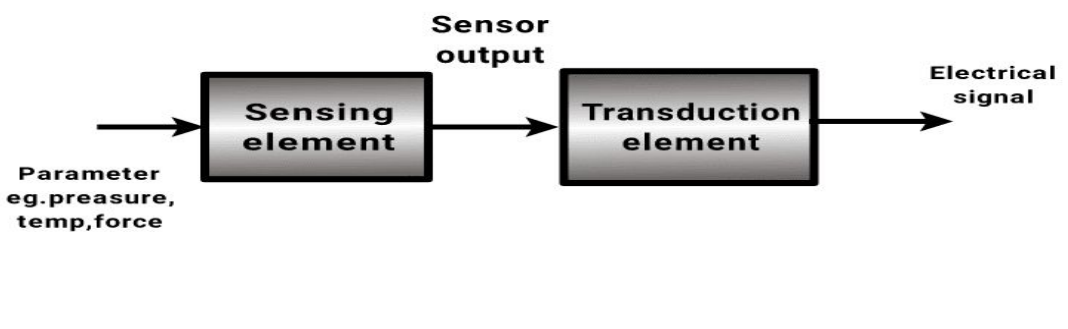


Fig. Transducer block diagram

Advantages of Transduction

The following are the advantages of converting the physical quantity into an electrical signal.

- The attenuation and amplification of the electrical signals are very easy.
- The electrical signal produces less friction error.
- The small power is required for controlling the electrical systems.
- The electrical signals are easily transmitted and processed for measurement.
- The component used for measuring the electrical signal is very compact and accurate.
- The electrical signals are used in telemetry.

Classification of transducers

Basically, there are two types of transducers: Electrical and Mechanical.

ELECTRICAL TRANSDUCER

An electrical transducer is a sensing device by which the physical, mechanical or optical quantity to be measured is transformed directly by a suitable mechanism into an electrical voltage/current proportional to the input measurand.

The electrical transducer converts the mechanical energy into an electric signal. The electrical signal may be voltage, current and frequency. The production of the signal depends on the resistive, inductive and capacitive effects of the physical input.

principles of sensing and transduction

An electrical transducer must have the following parameters

1. Linearity - The relationship between a physical parameter and the resulting electrical signal must be linear.

2. Sensitivity - This is defined as the electrical output per unit change in the physical parameter. High sensitivity is generally desirable for a transducer.

3. Dynamic Range - The operating range of the transducer should be wide, to permit its use under a wide range of measurement conditions

4. Repeatability - The input/output relationship for a transducer should be predictable over a long period of time. This ensures reliability of operation.

5. Physical Size - The transducer must have minimal weight and volume, so that its presence in the measurement system does not disturb the existing conditions.

Points to be considered for selecting a transducer

We have numerous Sensors / Transducers – it is better to select how to choose for our application

- **Principle of operation** – whether it is Resistive / Capacitive / Inductive / Optical Type
- **Availability** – Source of location, Availability in markets
- **Cost –** Moderate cost
- **Performance Figures:**
  - Characteristics
  - Speed
  - Response Time
  - Sensitivity
  - Hysteresis Effect

Classification of Transducer

Electrical transducers can be broadly classified into two major categories,

(i) Active, (ii) Passive.

- An active transducer generates an electrical signal directly in response to the physical parameter and does not require an external power source for its operation. Active transducers are self generating devices, which operate under energy conversion principle and generate an equivalent output signal
- Passive transducers operate under energy controlling principles, which makes it necessary to use an external electrical source with them. They depend upon the change in an electrical parameter (R, L and C).

Classification of Transducers

Transducers may be further classified into different categories depending upon the principle employed by their transduction elements to convert physical phenomena into output electrical signals.

The different electrical phenomena employed in the transduction elements of transducers are as follows.

1. Resistive
2. Inductive
3. Capacitive
4. Electro magnetic
5. Piezo-electric
6. Photo-emissive
7. Photo-resistive
8. Potentiometric
9. Thermo-electric
10. Frequency generating.

SENSOR TECHNOLOGY

E- Book Link

E-Book Name	Chapter	Page No.	Notes	URL
"Principles of Measurement Systems" by John P. Bentley	1,2,3,4	1 to 60		www.pearsoned.co.uk
Electronic instrumentation by H S Kalsi	1	1 to 30		http://www.mhhe.com/kalsi/ei3,